

MAHI AGARWAL

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SUMMARY

Final-year ECE undergraduate at IIIT Noida with strong programming fundamentals in C++ and Python, hands-on experience with data structures, competitive programming, and applied machine learning. Built multiple end-to-end systems spanning ML classification pipelines, secure real-time applications, and structured record-keeping tools. Seeking the Specialist Programmer role to apply strong problem-solving and coding skills to Infosys's AI-first engineering initiatives.

TECHNICAL SKILLS

- Languages & DSA: C++ (STL — vector, list, deque, competitive programming), Python, JavaScript, SQL
- Frontend (vanilla): HTML, CSS, JavaScript — DOM manipulation, localStorage, form validation
- Machine Learning: scikit-learn (SVM, KNN, Random Forest, MLP), Pandas, NumPy, feature engineering, model evaluation
- Databases & Tools: MySQL, Git/GitHub, VS Code, Excel, MATLAB, Cadence Virtuoso, HFSS, Proteus

PROJECTS

- Edible Oil Adulteration Detection — ML Classification Pipeline**

Python, scikit-learn, NumPy, Pandas

 - Built binary and multi-class Random Forest classifiers to detect and grade oil adulteration from microwave sensor (S11) frequency-response data.
 - Restructured the dataset to treat each frequency-S11 reading as an individual training sample rather than one sample per file, materially improving model accuracy.
- Audio Genre Classification**

Python, SVM, KNN, Random Forest, MLP

 - Trained and benchmarked four ML models (SVM, KNN, Random Forest, MLP) on the GTZAN audio dataset for genre classification as part of a IIIT ML PBL assignment; SVM achieved the best accuracy.
- SecureStream — Encrypted Webcam Streaming**

Python, OpenCV

 - Built a project that encrypts a live webcam feed in real time so the video stream can't be intercepted or read in transit.
 - Learned and applied core encryption concepts (key derivation, secure randomness, integrity checks) to keep the stream both private and tamper-proof.
- GirviLedger / GirwiBook — Digital Loan & Interest Record Systems**

JavaScript, HTML, CSS

 - Engineered an interest-calculation engine handling 30–365 day cycles with automated date-based computation, cutting manual calculation errors by 80–90%.
 - Implemented persistent offline storage, search/filter, and one-click Excel export for non-technical end users managing high volumes of records.
- Competitive Programming & DSA Practice**

C++, STL

 - Solved algorithmic problems using STL containers (vector, list, deque) with a focus on time/space complexity trade-offs.
 - Diagnosed and fixed a floating-point precision bug in a pow()-based Armstrong number check, replacing it with an integer-safe approach.

EDUCATION

Jaypee Institute of Information Technology (IIIT), Noida <i>B.Tech, Electronics and Communication Engineering</i>	July 2023 – Present
P.M.S. Public School, Moradabad <i>Class 12 (CBSE)</i>	Mar 2022 – Mar 2023

COURSES & CERTIFICATIONS

- AI Agents Intensive (Google/Kaggle) — Prompt engineering, agent workflows, GenAI prototyping
- Java (IIT Bhubaneswar) — OOP, control structures, exception handling, data structures